

Aggressive B-NHL — Burkitt & DLBCL Therapy



1. Burkitt: MYC, Ki-67 ~95%

WHAT YOU SEE

Burkitt express engine, MYC Ki67 95%

WHAT IT ENCODES

Burkitt lymphoma — MYC t(8;14), Ki-67 near 100%, fastest doubling time of any tumor. Starry-sky histology. Curable but demands intensive, dose-dense therapy.

BOARD TRAP

Standard R-CHOP undertreats Burkitt — its rapid growth needs intensive regimens (R-hyperCVAD or DA-EPOCH-R for select cases).

2. Shortest doubling time

WHAT YOU SEE

Speedometer <24h doubling time

WHAT IT ENCODES

Burkitt has a doubling time often <24-48h — the highest proliferative fraction of human tumors. This drives extreme tumor lysis risk and the need to start therapy urgently.

BOARD TRAP

The faster the doubling time, the higher the tumor lysis risk — Burkitt mandates aggressive rasburicase + hydration prophylaxis.

3. Endemic vs sporadic

WHAT YOU SEE

ENDEMIC / SPORADIC labels

WHAT IT ENCODES

Endemic Burkitt (Africa, EBV+, jaw/facial mass), sporadic (Western, abdominal/ileocecal mass), immunodeficiency-associated (HIV). All share MYC translocation and aggressive course.

BOARD TRAP

A jaw mass in an African child = endemic Burkitt (EBV+); an abdominal mass in a Western adult = sporadic — same biology, different presentation.

4. DA-EPOCH-R (BCL2-negative)

WHAT YOU SEE

DA-EPOCH-R car, BCL2 NEG

WHAT IT ENCODES

Dose-adjusted EPOCH-R is an option for Burkitt and for double-hit/high-grade B-cell lymphoma. Infusional dosing exploits proliferation kinetics; CNS prophylaxis is required.

BOARD TRAP

DA-EPOCH-R is favored in double-hit/high-grade lymphoma over R-CHOP, which underperforms in MYC-rearranged disease.

5. R-CHOP (DLBCL backbone)

WHAT YOU SEE

R-CHOP green car

WHAT IT ENCODES

R-CHOP (rituximab, cyclophosphamide, doxorubicin, vincristine, prednisone) is curative first-line for most DLBCL, ~6 cycles. ~60-70% cure overall depending on IPI.

BOARD TRAP

R-CHOP cures the majority of DLBCL — it is curative-intent, unlike palliative management of indolent FL.

6. Pola-R-CHP

WHAT YOU SEE

POLA-R-CHP car

WHAT IT ENCODES

Polatuzumab (anti-CD79b ADC) substituted for vincristine (R-CHP) improved progression-free survival in higher-risk DLBCL (POLARIX). Vincristine is dropped to avoid overlapping neuropathy.

BOARD TRAP

In Pola-R-CHP, vincristine (the 'O') is removed — adding it back would compound neurotoxicity.

7. CAR-T (relapsed DLBCL)

WHAT YOU SEE

CAR-T car, ZUMA-7

WHAT IT ENCODES

CD19 CAR-T (axi-cel/liso-cel) is standard for relapsed/refractory DLBCL. ZUMA-7/TRANSFORM showed superiority to salvage chemo + transplant in early relapse (<12 months).

BOARD TRAP

For early relapse (<12 mo) or primary-refractory DLBCL, CAR-T beats autologous transplant — don't default to ASCT salvage.

8. Relapse timing switch (12 mo)

WHAT YOU SEE

Relapse switch >12mo / <12mo, AUTO-SCT

WHAT IT ENCODES

Relapse timing dictates salvage: late relapse (>12 months) → salvage chemo + autologous stem cell transplant if chemosensitive; early/primary-refractory → CAR-T.

BOARD TRAP

Time-from-treatment is the fork — late chemosensitive relapse still favors ASCT; early relapse routes to CAR-T.

9. Response rate 80-90%

WHAT YOU SEE

80-90% banner

WHAT IT ENCODES

Aggressive curable B-NHL achieves high overall response rates with first-line immunochemotherapy. Interim/end PET (Deauville) assesses metabolic response.

BOARD TRAP

High response rate isn't the same as cure — a positive end-of-treatment PET (Deauville 4-5) signals residual disease needing salvage.

10. Salvage chemo

WHAT YOU SEE

SALVAGE car

WHAT IT ENCODES

Platinum-based salvage (R-ICE, R-DHAP) is used to re-induce chemosensitivity before autologous transplant in eligible relapsed DLBCL. Chemorefractory disease portends CAR-T.

BOARD TRAP

If salvage fails to achieve chemosensitivity, proceed to CAR-T rather than transplanting refractory disease.

11. R-EPOCH reinforced (high-grade)

WHAT YOU SEE

R-EPOCH reinforced steel car

WHAT IT ENCODES

Dose-intensified infusional regimens are reserved for double-hit/high-grade and CNS-risk disease, with intrathecal CNS prophylaxis. Reflects higher proliferative burden.

BOARD TRAP

Adding CNS prophylaxis matters in high-grade/MYC disease and Burkitt — omitting it risks CNS relapse.

12. Novel agents (POLA, BiTE, ADC)

WHAT YOU SEE

NOVEL AGENTS toolbox: POLA, bispecifics, ADC

WHAT IT ENCODES

Bispecific T-cell engagers (epcoritamab, glofitamab targeting CD20×CD3), antibody-drug conjugates (polatuzumab, loncastuximab), and tafasitamab+lenalidomide expand relapsed/refractory options.

BOARD TRAP

Bispecifics, like CAR-T, can cause cytokine release syndrome and ICANS — recognize fever/hypotension post-infusion as CRS, not sepsis alone.